

Writing Large Numbers in Scientific Notation

MA.8.NSO.1.4

I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

Yesterday the exponent laws finished — numbers and letters.

Today they go to work on twenty-digit numbers.

Warm-Up

On your own. Four minutes. No notes.

1. Rewrite with a positive exponent: $4x^{-3}$
2. Find the value: $(10^2)^3$
3. Find the value: $\sqrt{121}$
4. In 4,500,000, which place is the 4 in?

Warm-Up

Answers.

1. Rewrite with a positive exponent: $4x^{-3}$ **$4/x^3$**
2. Find the value: $(10^2)^3$ **$10^6 = 1,000,000$**
3. Find the value: $\sqrt{121}$ **11**
4. In 4,500,000, which place is the 4 in? **millions**

Notes

Copy the table.

I. Powers of ten name the places

10^1	10^2	10^3	10^4	10^5	10^6
10	100	1,000	10,000	100,000	1,000,000

The exponent counts the zeros — and names the place of the 1.

Notes

II. Scientific notation

$$a \times 10^n \quad \text{with} \quad 1 \leq a < 10$$

$$2,000,000 = 2 \times 10^6 \qquad 26,000,000 = 2.6 \times 10^7$$

The coefficient is at least 1 and less than 10: one nonzero digit before the decimal point.

The exponent is how many places the decimal point moves: 26,000,000. → 2.6, seven places, so 10^7 .

Notes

III. How many times larger?

$$\frac{6.4 \times 10^7}{3.2 \times 10^7} = 2 \qquad \frac{2 \times 10^5}{2 \times 10^2} = 10^3 = 1,000$$

$$\frac{8.4 \times 10^6}{2.1 \times 10^4} = 4 \times 10^2 = 400$$

Divide the coefficients. Divide the powers of 10. Multiply the two results.

On a screen, 8.4E6 means 8.4×10^6 .

Example 1

Standard form to scientific notation.

In 2018, funding for education in Florida totaled 30,100,000,000 dollars.

Express this value in scientific notation.

Example 1

Worked.

$$30,100,000,000 = 3.01 \times 10^{10} \quad \text{the 3 is in the ten-billions place}$$

$$3.01 \times 10,000,000,000 = 30,100,000,000 \quad \text{check}$$

Answer: 3.01×10^{10}

Your Turn

Same steps, your numbers. Boards up when done.

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

Your Turn

Answer.

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

the 1 is in the hundred-millions place

Answer: 1.105×10^8

Example 2

How many times larger?

How many times larger is 8.1×10^8 than 5.4×10^7 ?

Example 2

Worked.

$$\frac{8.1}{5.4} = 1.5 \quad \text{coefficients}$$

$$\frac{10^8}{10^7} = 10^1 = 10 \quad \text{powers of 10: subtract the exponents}$$

$$1.5 \times 10 = 15 \quad \text{multiply the two results}$$

Answer: 15 times larger

Your Turn

Same steps, your numbers. Boards up when done.

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

Your Turn

Answer.

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

$$0.75 \times 10 = 7.5$$

Answer: about 7.5 times larger

Whiteboards • Question 1 of 9

Boards up on three.

Write 4,000,000 in scientific notation.

Answer it.

Which place is the 4 in?

Whiteboards • Question 1 of 9

Answer.

Write 4,000,000 in scientific notation.

4 millions

Answer: 4×10^6

Whiteboards • Question 2 of 9

Boards up on three.

Write 7.2×10^4 in standard form.

Answer it.

Move the point four places.

Whiteboards · Question 2 of 9

Answer.

Write 7.2×10^4 in standard form.

$$7.2 \times 10,000$$

Answer: 72,000

Whiteboards • Question 3 of 9

Boards up on three.

A stadium's renovation used 356,000 bolts.

The stadium seats 71,000 people.

Write the number of bolts in scientific notation.

Answer it.

Every nonzero digit stays in the coefficient.

Whiteboards • Question 3 of 9

Answer.

A stadium's renovation used 356,000 bolts.

The stadium seats 71,000 people.

Write the number of bolts in scientific notation.

3.56 hundred-thousands

Answer: 3.56×10^5

Whiteboards • Question 4 of 9

Boards up on three.

Which is greater, 9.5×10^5 or 1.2×10^6 ?

Answer it.

Look at the powers first.

Whiteboards • Question 4 of 9

Answer.

Which is greater, 9.5×10^5 or 1.2×10^6 ?

950,000 versus 1,200,000

Answer: 1.2×10^6

Whiteboards • Question 5 of 9

Boards up on three.

Which is 52,000,000 written in scientific notation?

A. 5.2×10^7

B. 52×10^6

C. 5.2×10^6

D. 5.2×10^8

Answer it.

Whiteboards • Question 5 of 9

Answer.

Which is 52,000,000 written in scientific notation?

A. 5.2×10^7

B. 52×10^6

C. 5.2×10^6

D. 5.2×10^8

Answer: A — 5.2×10^7

Whiteboards • Question 6 of 9

Boards up on three.

How many times larger is 6×10^8 than 6×10^5 ?

A. 1,000

B. 3

C. 300

D. 1.6

Answer it.

Whiteboards • Question 6 of 9

Answer.

How many times larger is 6×10^8 than 6×10^5 ?

A. 1,000

B. 3

C. 300

D. 1.6

Answer: A — 1,000

Whiteboards • Question 7 of 9

Boards up on three.

How many times larger is 8.4×10^6 than 2.1×10^4 ?

Answer it.

Coefficients, then powers, then multiply.

Whiteboards • Question 7 of 9

Answer.

How many times larger is 8.4×10^6 than 2.1×10^4 ?

$$4 \times 10^2$$

Answer: 400

Whiteboards • Question 8 of 9

Boards up on three.

A computer displays a number as $3.5\text{E}9$.

Write the number in standard form.

Answer it.

E is not an error.

Whiteboards • Question 8 of 9

Answer.

A computer displays a number as 3.5E9.

Write the number in standard form.

$$3.5 \times 10^9$$

Answer: 3,500,000,000

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

Write your answer in sentences.

Use the definition.

Whiteboards · Question 9 of 9

Answer.

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

equal, yes — scientific notation, no

Answer: Equal, yes. Scientific notation, no: 20 is not less than 10. 2×10^6 .

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between standard and scientific notation: large numbers — 2AL

Compare large numbers — WEZ

Powers of ten — LJQ